1.

Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Ans 1

Subnet	Network Address	Subnet Mask	Broadcast Address
1	**192.168.10.0/26**	**255.255.255.192**	**192.168.10.63**
2	**192.168.10.64/26**	**255.255.255.192**	**192.168.10.127**
3	**192.168.10.128/26**	**255.255.255.192**	**192.168.10.191**
4	**192.168.10.192/26**	**255.255.255.192**	**192.168.10.255**

2.

For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

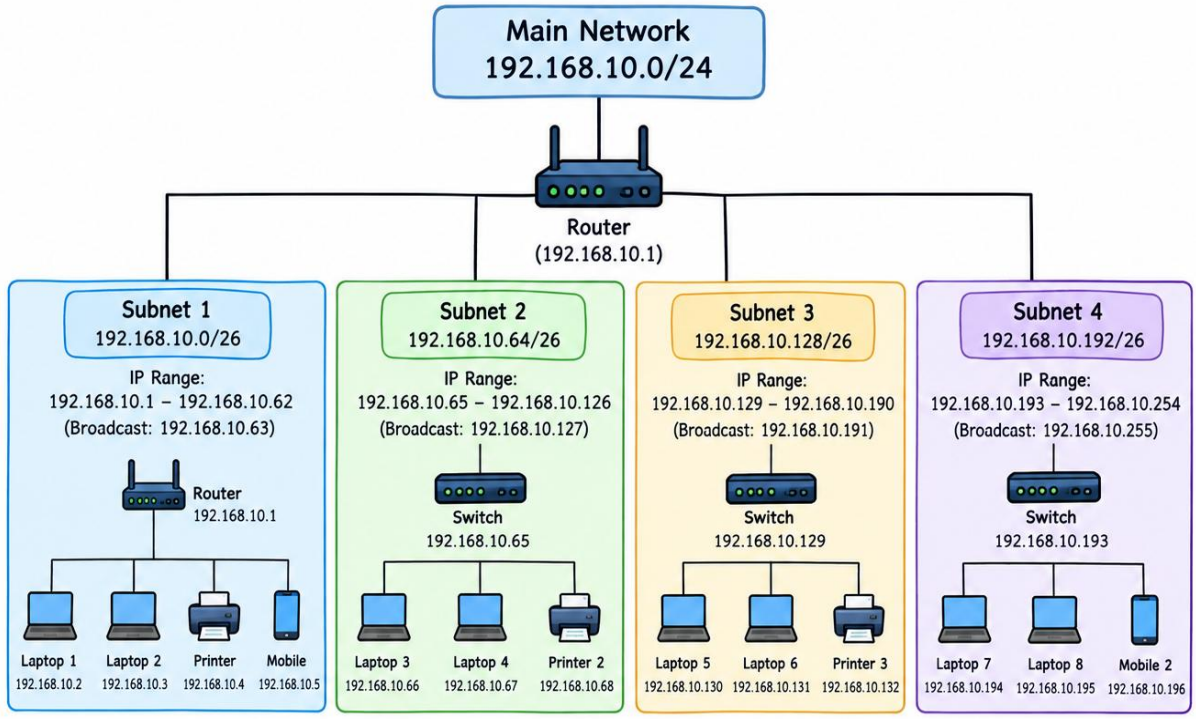
Hint: Make sure not to use the network or broadcast address for any device.

Ans 2

	Device	IP Address
.	Router	**192.168.10.1**
	Laptop 1	**192.168.10.2**
	Laptop 2	**192.168.10.3**
	Printer	**192.168.10.4**

3.

Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



Subnet Summary

Subnet	Subnet Address (/26)	Subnet Mask	Usable IP Range	Broadcast Address
Subnet 1	192.168.10.0/26	255.255.255.192	192.168.10.1 – 192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64/26	255.255.255.192	192.168.10.65 – 192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128/26	255.255.255.192	192.168.10.129 – 192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192/26	255.255.255.192	192.168.10.193 – 192.168.10.254	192.168.10.255

Device IP Assignment (Subnet 1)

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

Note: Network address and broadcast address are not assigned to any device.

4.

Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Ans 4

Short Step-by-Step

1. **Given network:** 192.168.10.0/24
2. **Create 4 subnets:**
   Borrow 2 bits because $\mathbf{2^2 = 4}$.
   New prefix = **/26**.
3. **Subnet mask:**
   255.255.255.192
4. **Hosts per subnet:**
   2^6 = **64 total IPs**
   64 − 2 = **62 usable IPs**
5. **Subnets:**
   - 192.168.10.0/26 → 1–62
   - 192.168.10.64/26 → 65–126
   - 192.168.10.128/26 → 129–190
   - 192.168.10.192/26 → 193–254
6. **Device assignment (Subnet 1):**
   - Router → 192.168.10.1
   - Laptop 1 → 192.168.10.2
   - Laptop 2 → 192.168.10.3
   - Printer → 192.168.10.4
   - Mobile → 192.168.10.5